

Suvra Nath, PhD

(DOB: 29.03.1989, Citizenship: Indian, Visa: German Blue Card)

Computational Neuroscientist, Multi-Omics & Translational Drug Discovery

Summary-Bio:

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Computational Neuroscientist with 16+ years of experience applying advanced analytics, multi-omics & machine learning to translational research and early-stage drug discovery. Proven track record in integrating complex biological and multimodal datasets to generate actionable insights supporting data-driven decision-making. Experienced in developing scalable analytical workflows, collaborating across cross-functional teams, and translating high-dimensional data into insights relevant for drug target & compound discovery with an intrinsically driven mission to improve *Mental Health & Brain Disorders*.

KEY SCIENTIFIC ACHIEVEMENTS

- Identified **10+ ligand–receptor interaction pairs** driving post-stroke regeneration, supporting translational discovery
- Developed ML-based multimodal imaging pipeline achieving **95% accuracy** in in vivo–ex vivo data integration
- Contributed to highly competitive **€200K+ grant funding** through data-driven insights and assay development
- Led cross-functional scientific research projects across **Asia, Europe & the USA**, managing teams of 3–10 scientists
- Awarded multiple **competitive fellowships/ scholarships** (*DFG-funded SFB-CRC/TRR-167, Women in-STEM, Justus & Louise van Effen Foundation, Merit-based Scholarships & Silver Medal*)

WORK EXPERIENCE(s) (contract/part-time positions not stated)

Senior Postdoctoral Scientist | Institute of Medical Psychology, Ludwigs–Maximilians–Universität, München, Germany | July 2024– Dec 2025 | Neural & Molecular Control of Sleep (*Grant Discontinued Abruptly!*)

Affiliation: Max Plank Institute of Biochemistry (Germany), Universitätsspital Bern (Switzerland) & Instituto de Neurociencias (Spain)

- Led multidisciplinary projects integrating proteomics, electrophysiology, and systems biology to identify sleep-related biomarkers and therapeutic targets
- Applied quantitative proteomics (*DIA/DDA/PTM*) and designed (*conceptual & computational*) foundation of ML-based analysis pipeline to uncover molecular drivers of neuronal activity during sleep
- Designed integrative workflows linking omics data to neural circuit function, supporting translational insights (*unfinished*)
- Increased experimental throughput by 30% through optimized data acquisition and pipeline integration (*unfinished*)
- Contributed to (writing, conceptualization & DoE) €200K+ grant proposals (DFG, CZI)
- Worked within cross-functional teams (3–5 members) on computational pipeline development using Agile methodologies

Postdoctoral Scientist | Department of Neuroscience, University of California San Diego, California, United States | August 2022–July 2024 | Neural Control of Obesity & Metabolism

Affiliation: Howard Hughes Medical Institute, Loudoun County, Virginia, United States

- Performed single-cell RNA-seq analysis to identify spatially resolved neuronal subtypes and candidate cell type markers
- Developed ML- pipeline (95% accuracy) for multimodal image registration and integration of behavior domain
- Integrated large-scale datasets (Allen Brain Atlas) for translational mapping of gene expression
- Developed and applied algorithms to process large-scale vol. images (HPC-based) for high-dimensional data analysis
- Delivered computational insights supporting very competitive NIH U01 (*High Risk & High Reward*) grant success

Graduate Researcher | Department of Neuroscience, University of Freiburg, Freiburg im Breisgau, Germany | May 2017–July 2022 | Neural Mechanism & Drug Target Discovery for Stroke

Affiliation: Charité Universitätsmedizin Berlin, Germany & Weizmann-Institut für Wissenschaften, Israel

- Discovered novel ligand–receptor interactions driving neuroregeneration using scRNA-seq and spatial analysis
- and successfully published 10+ novel ligand-receptor pairs mediating crosstalk between neural stem cells and myeloid cells during post-stroke regeneration, utilizing scRNA-seq transcriptomics, IHC (human & murine), in-situ hybridization, and advanced imaging techniques (*Light Sheet & Confocal*).
- Built reproducible (90%) experimental datasets from 50+ time-lapse imaging studies in rodent stroke models with controlled perturbations, enabling data-driven investigation of cellular mechanisms in post-stroke regeneration
- Built highly reproducible datasets (90%) from multimodal experiments enabling data-driven modeling
- Identified inflammatory pathways and biomarkers relevant to neurodegeneration and recovery
- Led and mentored 10+ researchers, fostering cross-functional collaboration

Research Associate | Department of Neuroscience, Erasmus Medical University, Rotterdam, The Netherlands | August 2015– January 2017 | Neural Mechanism Underlying Spinocerebellar Ataxia

Affiliation: Netherlands Institute for Neuroscience, The Netherlands and University of Edinburgh, Edinburgh, United Kingdom

- Proposed, led and successfully published an interdisciplinary project about the impact of chemically modified substrate induced mechanotransduction on cerebellar neurons
- Developed and analyzed multimodal datasets (molecular + electrophysiology) to model neuronal differentiation, identified translatable drug targets using human iPSC and animal models
- Coordinated interdisciplinary collaboration across 3 institutions and domains, awarded with highest honors for taking novel research initiative and outstanding academic performance

CORE RELEVANT SKILLS (selected skills; not exhaustive. Rapid learner with VERY strong adaptability)

Data Science & Machine Learning

- Python (NumPy, Pandas, SciPy, Scikit-learn, Napari), PyTorch, TensorFlow/Keras, R, MATLAB, Bash
- Supervised & unsupervised learning (classification, regression, shrinkage methods, parametric/non-parametric models, bias-variance trade-off), deep learning (CNN, U-Net), generative AI (VAE, scVI, scGen), transformer models, transfer learning, model evaluation (HPO, CV & explainability using SHAP scoring)
- Feature engineering, predictive modelling, dimensionality reduction, and high-dimensional data integration using large-scale biomedical databases (Allen Brain Atlas, Human Protein Atlas, Human Organoid Atlas, Connectivity Map, Drug Perturbation Database)
- Fine-tuning open-source transformer models (Hugging Face) and developing ML/AI-driven analytical workflows (basic knowledge)

Multi-Omics & Bioinformatics

- Single-cell & bulk RNA-seq, spatial omics (EASI-FISH, Seq-FISH), proteomics (DDA, DIA, PTM, SILAC), EEG
- Software-tools: Seurat, Scanpy, DESeq2, edgeR, limma, DIA-NN, MaxQuant, Perseus, Spectronaut, EEGLAB

Advanced Analytics & Statistical Modelling

- Predictive modelling, statistical inference, Bayesian & probabilistic modelling
- Frequentist (SPSS, GraphPad) and Bayesian approaches (PyMC3, latent variable models, MCMC, variational inference)
- Experimental design, hypothesis testing, model validation, and data-driven decision-making

Software, Data Engineering & Cloud

- Python (OOP, intermediate developer knowledge), (Ann/) SQL Alchemy, NoSQL (Mongo DB)
- AWS (S3, Redshift, SageMaker), GCP/BigQuery (limited user experience), HPC (Slurm, IBM Load Balancer)
- Nextflow, SnapLogic (testing version), GitLab/GitHub Actions, familiarity with FAIR data principles
- Docker, Flask/Fast API, backend architecture, scalable full-stack software system design (actively learning)
- Retrieval-Augmented Generation (RAG) pipeline: LlamaIndex, Vector DB (basic)
- LLM APIs (Open AI, Gemini & Anthropic), Prompt Engineering/ AI code agents (Codex, Claude) within IDE (VS Studio, PyCharm, Antigravity, Cursor)

Multimodal Data Integration & Digital Health

- Integration of imaging, transcriptomics, proteomics, spatial omics & SysBio datasets (e.g. latent factor integration)
- Image analysis, stitching (Apache-Spark), segmentation (Starfinity), registration (Bigstream), visualization (Plotly-Dash)

Life Sciences, Drug Discovery & Scientific Innovation

- DL/AI-enabled drug compound/target identification, and drug repurposing (AlphaFold2, PyMOL)
- Strong interest in applying advanced analytics to experimental medicine and clinical research datasets

Selected Outputs & Additional Multidomain Expertise

- [7 publications](#) (3 ongoing & under revision) and >130 citations
- **Global Scientific Outreach:** 30+ invited scientific talks (international conferences)
- **Languages:** English, German (B2 ongoing), Hindi, Bengali, Japanese (basic)
- **Relevant Skills/ Certifications:** JIRA, Mattermost, Confluence, Trello, DAMA (ongoing), SCRUM [master](#) & [product owner](#), [Six Sigma](#), [GCP](#), [GMP](#), [Drug Discovery \(Conceptualization till Clinical Trial Phase\)](#), Regulatory Affairs Framework EU & FDA (ongoing), EU Scientific Patent/ Trademark & Design-IP Rights

EDUCATION

- [Micro Master of Business Administration \(STEM\)](#) Rady School of Management-UCSD, California, United States • 2023
- **Ph.D. in Neuroscience** (*summa cum laude*) University of Freiburg, Germany • 2022
- **MS in Biomedical (Neural Tissue) Engineering** (*magna cum laude, honors*) TU Delft • The Netherlands • 2016
- **BTech in Biomedical (Brain Signal Processing) Engineering** (*summa cum laude*) SRM University • India • 2013